

D378

Digital Marketing Science

Armored Drill

8 concepts · 12 items · 4 transfer items

What this is. Every concept in this course's public description, re-asked with more disguise than a practice test uses. Each stem carries facts that do not decide it. Every wrong answer is a named trap.

Source. Scope from the course's public description in the WGU Institutional Catalog (September 2026); items written from the field's standard body of knowledge. No WGU assessment item, course material or proprietary content is used or reproduced. Not affiliated with or endorsed by Western Governors University.

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How to run it

Paper first. Pencil. One pass, no notes, a timer if you want the pressure.

1. **Before you read the options**, fill the strip under the item: the target word (what is actually being asked), and the one fact in the stem that decides it.
2. **Principle before arithmetic**. Name the rule, then compute.
3. **Camouflage**. Most of every stem does not matter. Ask: which single fact decides this?
4. **Inverse pairs are the bait**. If two options are opposites, one of them is the trap built for you.
5. **Mark sure, unsure or guess** honestly. Sure-and-wrong is the most valuable miss on the page — it is a wrong mental model.
6. **Absolutes** (always, never, guarantee) are usually wrong. And recognizing an answer is not the same as owning it.
7. Items with a **heavy black bar** on the left are **transfer items**: the same principle, reversed or solved for a different variable.
8. Stop at the STOP page. Then grade, fill the triage grid, and tally the traps you fell for. The biggest tally is what you drill next.

The strip, filled in, looks like this:

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	_____	

Items

1.

An outdoor-gear brand spent \$10,000 on shopping ads that brought in \$40,000 in revenue. Its products carry a 20% profit margin before ad costs. The brand's photographer is based in Denver. Which statement is correct?

- A. ROAS is 4:1, but the campaign loses \$2,000 after ad costs.
- B. ROAS is 3:1 because \$30,000 is left after ad spend.
- C. ROAS is 0.25:1.
- D. ROAS is 4:1, so the campaign is profitable.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

2. [TRANSFER — reversed stem / different variable]

A brand earns a 25% margin on sales. What is the minimum ROAS it needs to break even on ad spend?

- A. 1:1
- B. 0.25:1
- C. 4:1
- D. 1.25:1

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

3.

A podcast network charges a \$25 CPM. A kitchen-gadget brand buys 400,000 impressions for a holiday flight; the host has a devoted audience of home cooks. What does the buy cost?

- A. \$16,000
- B. \$25
- C. \$10,000
- D. \$10,000,000

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

4. [TRANSFER — reversed stem / different variable]

An advertiser paid \$6,000 for a display campaign at a \$12 CPM. How many impressions did it buy?

- A.** 72,000
- B.** \$0.012 per impression
- C.** 500
- D.** 500,000

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

5.

A newsletter was sent to 20,000 addresses and 1,000 bounced. It was opened 4,750 times and clicked 380 times. The subject line used an emoji for the first time. What is the click-to-open rate?

- A.** 25%
- B.** 1.9%
- C.** 8%
- D.** 2%

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

6.

After two days, version B of a checkout page shows a 12% lift with 80 visitors per version, and the testing tool reports 62% confidence. The design team loves B's new font. The plan called for 95% confidence. What should the analyst do?

- A.** Scrap it; tests are never valid under 10,000 visitors.
- B.** Keep the test running until the planned sample and 95% confidence are reached.
- C.** Pick B, since the team prefers its font.
- D.** Declare B the winner based on the 12% lift.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

7. [TRANSFER — reversed stem / different variable]

A test returns a p-value of 0.03 against a significance threshold of 0.05. How should the result be read?

- A.** Statistically significant: a difference this large would be unlikely if the versions truly performed the same.
- B.** B will always outperform A from now on.
- C.** Not significant, because 0.03 is smaller than 0.05.
- D.** There is a 97% chance that B is better.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

8.

A shopper first found a hiking-boot brand through a blog post written by a guest author, later clicked a Facebook ad, and finally bought \$200 of boots after clicking a branded search ad. Under a linear attribution model, how much credit does the Facebook ad get?

- A.** \$0
- B.** \$66.67
- C.** \$100
- D.** \$200

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

9. [TRANSFER — reversed stem / different variable]

Which attribution model gives 100% of the credit to the touchpoint that first introduced the customer to the brand?

- A.** First-touch
- B.** Linear
- C.** Last-touch
- D.** Time-decay

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

10.

An online tea shop gets 30,000 visits a month, converts 2.5% of them, and has an average order value of \$40. The owner is considering a new logo. What is monthly revenue?

- A.** \$3,000,000
- B.** 750
- C.** \$1,200,000
- D.** \$30,000

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

11.

A subscription box costs \$30 a month with a 40% gross margin, and the average customer stays 15 months. It costs \$120 to acquire a customer. The company also sponsors a minor-league baseball team. What is the margin-based CLV:CAC ratio?

- A.** 1.5:1, which is healthy because any ratio above 1:1 always is
- B.** 1.5:1, below the common 3:1 benchmark
- C.** 3.75:1
- D.** 0.67:1

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

12.

In a search auction, Advertiser A bids \$4.00 with a Quality Score of 5, and Advertiser B bids \$2.50 with a Quality Score of 9. B's agency is based in London. Using $\text{Ad Rank} = \text{bid} \times \text{Quality Score}$, who ranks higher?

- A.** A, because it bids more
- B.** A, since the highest bidder always wins any auction
- C.** A, because a lower Ad Rank wins
- D.** B, with an Ad Rank of 22.5 vs. 20

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

STOP

Answers and dissection follow. Finish every item and every strip first.

Key and dissection

Struck-through text is the camouflage: facts that do not decide the item.

1. Answer A Concept 1

An outdoor-gear brand spent \$10,000 on shopping ads that brought in \$40,000 in revenue. Its products carry a 20% profit margin before ad costs. ~~The brand's photographer is based in Denver.~~ Which statement is correct?

Target word: ROAS and profitability **Decides:** Margin profit \$8,000 is less than \$10,000 ad spend.

A	Correct	$\$40,000 / \$10,000 = 4:1$; $\$40,000 \times 20\% = \$8,000 - \$10,000 = -\$2,000$.
B	Confusion set	Mixes a revenue-based return figure with ROAS.
C	Inverted operation	Divides spend by revenue.
D	Dropped element	Ignores the 20% margin; break-even ROAS here is 5:1.

2. Answer C Concept 1 · transfer

A brand earns a 25% margin on sales. What is the minimum ROAS it needs to break even on ad spend?

Target word: solve for break-even ROAS **Decides:** Break-even ROAS = $1 / \text{margin}$.

A	Dropped element	Ignores margin; revenue equal to spend loses money.
B	Wrong stem number	Uses the margin itself as the ratio.
C	Correct	$1 / 0.25 = 4$; $\$4 \text{ revenue} \times 25\% = \$1 \text{ margin per } \$1 \text{ of ads}$.
D	Inverted operation	Adds the margin to 1 instead of dividing 1 by it.

3. Answer C Concept 2

A podcast network charges a \$25 CPM. A kitchen-gadget brand buys 400,000 impressions for a holiday flight; ~~the host has a devoted audience of home cooks.~~ What does the buy cost?

Target word: total cost **Decides:** $400 \text{ thousands} \times \25 .

A	Inverted operation	$400,000 / 25$ divides instead of applying the rate per thousand.
B	Wrong stem number	The rate, not the cost.
C	Correct	$(400,000 / 1,000) \times \$25 = \$10,000$.
D	Dropped element	Forgets to divide impressions by 1,000.

4. Answer D Concept 2 · transfer

An advertiser paid \$6,000 for a display campaign at a \$12 CPM. How many impressions did it buy?

Target word: solve for impressions **Decides:** Impressions = $(\text{cost} / \text{CPM}) \times 1,000$.

A	Inverted operation	Multiplies cost by the rate.
B	Adjacent category	Cost per single impression, not the count.
C	Dropped element	Forgets the "per thousand."
D	Correct	$\$6,000 / \$12 = 500 \text{ thousand impressions}$.

5. Answer C Concept 3

A newsletter was sent to 20,000 addresses and 1,000 bounced. It was opened 4,750 times and clicked 380 times. ~~The subject line used an emoji for the first time.~~ What is the click-to-open rate?

Target word: CTOR **Decides:** Clicks / opens = 380 / 4,750.

A	Adjacent category	4,750 / 19,000 is the open rate.
B	Dropped element	380 / 20,000 uses sent and drops both bounces and opens.
C	Correct	380 / 4,750 = 0.08.
D	Confusion set	380 / 19,000 delivered is the click-through rate, not CTOR.

6. Answer B Concept 4

After two days, version B of a checkout page shows a 12% lift with 80 visitors per version, and the testing tool reports 62% confidence. ~~The design team loves B's new font.~~ The plan called for 95% confidence. What should the analyst do?

Target word: when to call the test **Decides:** 62% confidence and a tiny sample fall short of the 95% plan.

A	Absolute language	Required sample depends on the effect size; there is no universal cutoff.
B	Correct	Early lifts on small samples are unreliable.
C	Camouflage	Opinion does not decide a test.
D	Dropped element	Ignores the significance requirement.

7. Answer A Concept 4 · transfer

A test returns a p-value of 0.03 against a significance threshold of 0.05. How should the result be read?

Target word: interpret p-value **Decides:** 0.03 is below 0.05.

A	Correct	p below the threshold means reject the no-difference assumption.
B	Absolute language	Significance does not guarantee future results.
C	Inverted operation	Smaller p is stronger evidence, not weaker.
D	Confusion set	A common misreading; p is not the probability the hypothesis is true.

8. Answer B Concept 5

A shopper first found a hiking-boot brand through a blog post ~~written by a guest author~~, later clicked a Facebook ad, and finally bought \$200 of boots after clicking a branded search ad. Under a linear attribution model, how much credit does the Facebook ad get?

Target word: linear credit for one touch **Decides:** Three touchpoints share \$200 equally.

A	Confusion set	What Facebook gets under first-touch or last-touch, not linear.
B	Correct	\$200 / 3 touchpoints.
C	Dropped element	Splits credit between the two ads and drops the blog touchpoint.
D	Wrong stem number	The full order value.

9. Answer A Concept 5 • transfer

Which attribution model gives 100% of the credit to the touchpoint that first introduced the customer to the brand?

Target word: reverse: from rule to model **Decides:** All credit to the first interaction.

A	Correct	Credits only the first interaction.
B	Confusion set	Spreads credit evenly.
C	Inverted operation	Credits only the final interaction.
D	Adjacent category	Gives the most credit to touches nearest the sale.

10. Answer D Concept 6

An online tea shop gets 30,000 visits a month, converts 2.5% of them, and has an average order value of \$40. ~~The owner is considering a new logo.~~ What is monthly revenue?

Target word: monthly revenue **Decides:** $30,000 \times 0.025 \times \40 .

A	Wrong stem number	Uses 2.5 instead of 0.025.
B	Confusion set	The number of orders, not revenue.
C	Dropped element	Leaves out the conversion rate.
D	Correct	$750 \text{ orders} \times \$40 = \$30,000$.

11. Answer B Concept 7

A subscription box costs \$30 a month with a 40% gross margin, and the average customer stays 15 months. It costs \$120 to acquire a customer. ~~The company also sponsors a minor league baseball team.~~ What is the margin-based CLV:CAC ratio?

Target word: CLV:CAC **Decides:** $\text{CLV} = \$30 \times 0.40 \times 15 = \180 .

A	Absolute language	Right number, wrong judgment; 1.5:1 leaves little room for overhead.
B	Correct	$\$180 / \$120 = 1.5$.
C	Dropped element	Uses revenue CLV (\$450) and drops the margin.
D	Inverted operation	Divides CAC by CLV.

12. Answer D Concept 8

In a search auction, Advertiser A bids \$4.00 with a Quality Score of 5, and Advertiser B bids \$2.50 with a Quality Score of 9. ~~B's agency is based in London.~~ Using Ad Rank = bid × Quality Score, who ranks higher?

Target word: who ranks higher **Decides:** $A = 20$; $B = 22.5$.

A	Dropped element	Ignores Quality Score.
B	Absolute language	Search auctions weigh relevance, not just bid.
C	Inverted operation	The higher Ad Rank wins the better position.
D	Correct	$\$2.50 \times 9 = 22.5$ beats $\$4.00 \times 5 = 20$.

Triage grid

Write each item number in the box it belongs to.

	Right	Wrong
Sure	Owned.	Concept inversion — top priority. A wrong mental model; it survives rewording.
Unsure / guess	Recognition, not mastery. Re-drill with the transfer.	Not learned yet.

Trap tally

For every miss, find the letter you chose in the key and tally its trap. The biggest number is the pattern to drill.

Trap	What it does	In this drill	Your misses
Dropped element	Leaves out a required term or step.	10	
Wrong stem number	A figure copied from the prompt that is not the answer.	4	
Adjacent category	The right idea in the wrong bucket.	3	
Inverted operation	Direction flipped: add/subtract, cause/effect, higher/lower.	8	
Absolute language	Always, never, guarantee — where judgment applies.	4	
Confusion set	Every option is real; the test is exact role-matching.	6	
Camouflage	Answers a stem fact that was never the question.	1	

Anchor rows — the concept list

One row per concept: the rule to lock in, the trap that catches people, and the tell in the question that signals it.

#	Concept	Rule	Trap	Tell
1	ROAS vs. profit	ROAS = revenue / ad spend; whether that is profitable depends on margin, and break-even ROAS = 1 / margin.	Assuming any ROAS above 1:1 is profitable.	Revenue, ad spend, and a margin percentage together.
2	CPM math	Cost = (impressions / 1,000) × CPM.	Forgetting the “per thousand.”	A CPM rate and an impression count.
3	Email engagement ratios	Delivered = sent – bounces; open rate and CTR use delivered as the base; click-to-open rate (CTOR) = clicks / opens.	Mixing up CTR and CTOR, or using sent instead of delivered.	Sent, bounced, opened, and clicked all listed.
4	Statistical significance in testing	Call a test only after the planned sample is reached and the result clears the significance threshold (commonly $p < 0.05$).	Declaring a winner from an early lift or reading $p < 0.05$ as not significant.	A lift figure with a small sample or a confidence level.
5	Attribution models	First-touch credits the first interaction, last-touch the final one, linear splits credit evenly across all touchpoints.	Applying last-touch logic when a different model is named.	A multi-step customer path plus a named model.
6	Funnel revenue math	Revenue = visitors × conversion rate × average order value.	Dropping a step or using a percent as a whole number.	Traffic, conversion rate, and order value in one stem.
7	Customer lifetime value vs. acquisition cost	Margin-based CLV = revenue per period × margin × periods retained; a CLV:CAC of about 3:1 is a common health benchmark.	Using revenue instead of margin, which inflates CLV.	Monthly price, margin, retention, and CAC together.
8	Ad Rank and Quality Score	In a simplified search auction, Ad Rank = max bid × Quality Score, so a relevant ad can outrank a higher bid.	Assuming the highest bidder always wins the top spot.	Two advertisers with different bids and Quality Scores.

Course scope (public description): Digital Marketing Science provides learners with knowledge in the dynamic and evolving field of marketing that uses the internet and digital technologies to reach customers. This course provides foundational knowledge in digital marketing disciplines, including social media marketing, email marketing, content marketing, marketing automation, search engine optimization (SEO), paid search (PPC), conversion rate optimization (CRO), mobile marketing, and web analytics. Learners explore how a digital marketing strategy is developed and executed, along with how digital marketing is integrated within an organization's overall strategy. This course provides learners with the opportunity to complete the HubSpot Marketing Software certification to learn marketing automation software that is used in industry to assist marketers in their digital marketing execution. Emphasis throughout the course is placed on industry best practices, and the course content aligns with the Online Marketing Certified Associate (OMCA) certification to prepare students for this third-party certification.